MARKING SCHEME

Model Paper — 03

Third Term Test

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Note (for teachers):

- Part I – A: 2 marks per question (normally 1 mark for the correct method, 1 mark for the correct final answer; where there is no working, 1 mark for the answer and 1 for the method or reason shown). Total 30 marks.
- Part I – B: 10 marks per question. Total 20 marks. The marks for each step are given in the right-hand column.
- Part II: candidates answer any **five** questions; 10 marks per question. Total 50 marks. The marks for each step are given in the right-hand column.
- Give full marks for correct alternative methods. Follow-through marks may be given for a correct method applied to a wrong earlier answer. Accept a reasonable range for measured values and values read from a graph.

PART I – A — Answers

1. $110110_2 = 1\times2^5 + 1\times2^4 + 0\times2^3 + 1\times2^2 + 1\times2^1 + 0\times2^0 = 32 + 16 + 4 + 2$

Final answer: 54

(2 marks)

2. 4th term $= 4^2 + 1 = 17$; 5th term $= 5^2 + 1 = 26$

Final answer: 17 and 26

(2 marks)

3. $\frac{5}{3} + \frac{11}{4} - \frac{3}{2} = \frac{20 + 33 - 18}{12} = \frac{35}{12}$

Final answer: $2\frac{11}{12}$

(2 marks)

4. 1 book = 960 ÷ 4 = Rs 240 ; 7 books = 240 × 7

Final answer: Rs 1 680

(2 marks)

5. $4x^2 - 25 = (2x)^2 - 5^2$ (difference of two squares)

Final answer: $(2x - 5)(2x + 5)$

(2 marks)

6. $x^{-2+5} \times 1$ ($y^0 = 1$)

Final answer: x^3

(2 marks)

7. 0.04572: the first significant figure is 4 (the zeros before it are not significant); the 3rd significant figure is 5, so round up 0.045 → 0.046

Final answer: 0.046

(2 marks)

8. $\frac{x}{3} = 7 - 2 = 5 \Rightarrow x = 5 \times 3$

Final answer: x = 15

(2 marks)

9. AC = AB + BC ; BD = BC + CD ; AB = CD (given). Equals (AB, CD) added to the same thing (BC) give equal sums.

Final answer: AC = BD ; Axiom: when equals are added to equals, the sums are equal

(2 marks)

10. Sum of ratio parts = 2 + 3 + 4 = 9 ; largest angle = $\frac{4}{9} \times 180°$

Final answer: 80°

(2 marks)

11. Perimeter = $\pi r + 2r = \frac{22}{7} \times 14 + 2 \times 14 = 44 + 28$

Final answer: 72 cm

(2 marks)

12. The hypotenuse is AC = 17 cm, AB = 8 cm. $BC^2 = 17^2 - 8^2 = 289 - 64 = 225$

Final answer: BC = 15 cm

(2 marks)

13. $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 9 - 4$

Final answer: 17

(2 marks)

14. Interior angle = $\frac{(10 - 2) \times 180°}{10} = \frac{1440°}{10}$ (or $180° - 360° \div 10 = 180° - 36°$)

Final answer: 144°

(2 marks)

15. 6 litres = 6 000 ml ; 6 000 ÷ 750

Final answer: 8 bottles

(2 marks)

PART I – B — Answers and mark allocation

Question 1

(i)	
Figure 1 = 4 ; Figure 2 = 7 (both correct)	**1**
Figure 3 = 10	**1**
(ii)	
Each figure needs 3 more sticks than the previous one (common difference 3): 4, 7, 10, …	**1**
$T_n = 3n + 1$ (equivalently $4 + 3(n - 1)$)	**1**
(iii)	
$T_{25} = 3(25) + 1$	**1**
= 76 matchsticks	**1**
(iv)	
$3n + 1 = 61 \Rightarrow 3n = 60$	**1**
$n = 20$, i.e. the 20th figure (check: $3 \times 20 + 1 = 61$)	**1**
(v)	
$3n + 1 = 100 \Rightarrow 3n = 99 \Rightarrow n = 33$ (a whole number)	**1**
Yes — the 33rd figure uses exactly 100 matchsticks (check: $3 \times 33 + 1 = 100$)	**1**

Total marks: 10

Question 2

(i) Angle = $\frac{\text{amount}}{60\,000} \times 360°$	
Food: $\frac{24\,000}{60\,000} \times 360° = 144°$	**1**
Education: $\frac{12\,000}{60\,000} \times 360° = 72°$	**1**
Transport: $\frac{9\,000}{60\,000} \times 360° = 54°$	**1**
Other: $\frac{15\,000}{60\,000} \times 360° = 90°$ (check: 144 + 72 + 54 + 90 = 360)	**1**
(ii) A correct pie chart shows a circle of radius 3 cm divided into four sectors of 144°, 72°, 54° and 90°	
Circle of radius 3 cm drawn with compasses, centre marked	**1**
Sectors drawn with the correct angles (±2° accepted), using a protractor	**1**
Each sector labelled (Food, Education, Transport, Other, with angles or amounts)	**1**
(iii)	
Percentage = $\frac{9\,000}{60\,000} \times 100$	**1**
= $\frac{3}{20} \times 100$ (or 54° ÷ 360° × 100)	**1**
= 15 %	**1**

Total marks: 10

PART II — Answers and mark allocation

Question 1

(a)(i)	
Total cost = 200 × 60 = Rs 12 000	**1**
Mangoes sold = 200 − 20 = 180 ; total income = 180 × 80 = Rs 14 400	**1**
(a)(ii)	
Profit = 14 400 − 12 000 = Rs 2 400	**1**
Profit percentage = $\frac{2\,400}{12\,000} \times 100$	**1**
= 20 %	**1**
(b)(i)	
150 000 000 = 1.5 × 100 000 000	**1**
= 1.5×10^8 km	**1**
(b)(ii)	
Time = distance ÷ speed	**1**
= $\frac{1.5 \times 10^8}{3 \times 10^5} = 0.5 \times 10^3$	**1**
= 500 seconds	**1**

Total marks: 10

Question 2

(a)	
(i) $(3a^2b)^2 = 3^2 \times a^4 \times b^2$	**1**
$= 9a^4b^2$	**1**
(ii) $2^{5+(-3)-(-1)} = 2^3$	**1**
$= 8$	**1**
(iii) $(x^3y^{-2})^0 = 1$; $1 \times 5 = 5$	**1**
(b)	
(i) $2^x = 2^5 \Rightarrow x = 5$	**1**
(ii) $3^{x+1} = 3^4 \Rightarrow x + 1 = 4$	**1**
$x = 3$	**1**
(c)	
$(5.2 + 4.8) \times 10^3 = 10 \times 10^3$	**1**
$= 1 \times 10^4$ (scientific notation)	**1**

Total marks: 10

Question 3

(a)	
$2x - 2 > x + 3 \Rightarrow x > 5$	**1**
Number line: open (unfilled) circle at 5 and an arrow/shading to the right	**1**
Smallest integer = 6	**1**
(b)(i)	
$x = 0 \Rightarrow y = 3$; $x = 2 \Rightarrow y = -1$ (both correct; check with the table: y decreases by 2 each time)	**1**
Suitable axes and scale; points (−1, 5), (0, 3), (1, 1), (2, −1), (3, −3) plotted correctly	**1**
A straight line drawn through the points (sloping down from left to right, cutting the y-axis at 3 and the x-axis at 1.5)	**2**
(b)(ii)	
Gradient = −2	**1**
Intercept (on the y-axis) = 3	**1**
(b)(iii)	
When x = 4, y = −2(4) + 3 = −5, so (4, −5) lies on the line	**1**

Total marks: 10

Question 4

(a)	
From $x - y = 2 : x = y + 2$ (or multiply the second equation by 2 to eliminate y)	**1**
$3(y + 2) + 2y = 16 \Rightarrow 5y + 6 = 16$	**1**
$y = 2$	**1**
$x = 4$ (check: $3(4) + 2(2) = 16$ and $4 - 2 = 2$)	**1**
(b)	
$3(2x - 1) = 15 \Rightarrow 2x - 1 = 5$	**1**
$2x = 6 \Rightarrow x = 3$ (check: $3(5) \div 5 = 3$)	**1**
(c)	
(i) $C = \frac{5(95 - 32)}{9} = \frac{5 \times 63}{9}$	**1**
$C = 35$ (°C)	**1**
(ii) $9C = 5(F - 32) \Rightarrow F - 32 = \frac{9C}{5}$	**1**
$F = \frac{9C}{5} + 32$	**1**

Total marks: 10

Question 5

(a)	
Draw a line through C parallel to AB (so it is also parallel to DE), and extend it to the left of C	**1**
The angle between CA and this line = $B\hat{A}C = 50°$ (alternate angles, AB ∥ the line)	**1**
The angle between CD and this line = $C\hat{D}E = 30°$ (alternate angles, DE ∥ the line)	**1**
$A\hat{C}D = 50° + 30° = 80°$	**1**
(b)	
$PQ = PR \Rightarrow P\hat{R}Q = P\hat{Q}R = 65°$ (base angles of an isosceles triangle)	**1**
$Q\hat{P}R = 180° - (65° + 65°) = 50°$	**1**
(c)	
Let the exterior angle = x ; interior angle = 5x ; $5x + x = 180°$ (angles on a straight line)	**1**
$6x = 180° \Rightarrow$ exterior angle = 30°	**1**
Number of sides = $360° \div 30°$	**1**
= 12 (check: interior angle $150° = 5 \times 30°$)	**1**

Total marks: 10

Question 6

(i)	
PQ = 8 cm drawn accurately (±1 mm)	**1**
(ii) Perpendicular at P (any standard method, e.g. arcs of equal radius cut PQ extended and PQ, then arcs of larger equal radius meet above P)	
PQ extended on the other side of P, and an arc centred P cutting both sides at equal distances	**1**
Two equal arcs (larger radius) from these two points intersecting above P	**1**
Line from P through the intersection point; the angle is 90° (construction arcs shown)	**1**
(iii)	
R marked on the perpendicular at 6 cm from P (±1 mm) and QR joined to complete triangle PQR	**1**
(iv)	
Measured QR ≈ 10 cm (9.9 – 10.1 cm accepted)	**1**
$QR^2 = PQ^2 + PR^2 = 8^2 + 6^2 = 64 + 36 = 100 \Rightarrow QR = 10$ cm	**1**
(v)	
Equal arcs (radius greater than half of QR) from Q and R, intersecting on both sides of QR	**1**
Line through the two intersections drawn; it meets QR at M, the mid-point of QR (QM = MR ≈ 5 cm)	**1**
PM measured ≈ 5 cm (4.9 – 5.1 cm accepted) (M is the centre of the circle through P, Q, R, so PM = ½ QR = 5 cm)	**1**

Total marks: 10

Question 7

(a)(i)	
Area = $\frac{1}{2} \times (30 + 20) \times 16$	**1**
$= 400\ \text{m}^2$	**1**
(a)(ii)	
Cost = 400×125	**1**
= Rs 50 000	**1**
(b)(i)	
Sample space = {1, 2, 3, 4, 5, 6}	**1**
(b)(ii)	
Numbers greater than 4: 5, 6 (2 outcomes)	**1**
$P = \frac{2}{6} = \frac{1}{3}$	**1**
(b)(iii)	
Factors of 6 on the die: 1, 2, 3, 6 (4 outcomes)	**1**
$P = \frac{4}{6} = \frac{2}{3}$	**1**
(b)(iv)	
Primes: 2, 3, 5 ; not prime: 1, 4, 6 $\Rightarrow P = \frac{3}{6} = \frac{1}{2}$	**1**

Total marks: 10